A Concrete Example of the Zakharov–Shabat Construction

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In this note we give a concrete example of the Zakharov–Shabat construction, and verify that it indeed produces an integrable system. For the Zakharov–Shabat construction one may consult the relevant part of [1].

1 An example

Consider the simplest case, in which there is only a first-order pole at 0. Write $L = L_0 + \frac{1}{\lambda}L_1$. Suppose $P = \lambda L^2$; a direct computation gives

$$M = (P(L))_- = \frac{1}{\lambda}L_1^2,$$

$$\frac{dL}{dt} = [L, M] = \frac{1}{\lambda}[L_0, L_1^2],$$

that is,

$$\begin{aligned} \dot{L_0} &= 0, \\ \dot{L_1} &= [L_0, L_1^2]. \end{aligned}$$

Hence L_0 is constant, and we diagonalize it:

$$L_0 = \begin{pmatrix} u & 0 \\ 0 & v \end{pmatrix}.$$

Write

$$L_1 = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

Expanding further,

$$\begin{aligned} \dot{a} &= 0, \\ \dot{d} &= 0, \\ \dot{b} &= (a+d)(u-v)b, \\ \dot{c} &= (a+d)(v-u)c. \end{aligned}$$

In the end this reduces to the differential equation

$$\dot{x} = x,$$
$$\dot{y} = -y.$$

The corresponding symplectic form is $dx \wedge dy$, and the Hamiltonian is xy. Note that this differs from a general Liouville integrable system, where one usually assumes that the Hamiltonian grows in every direction (i.e. that $\{H \leq \epsilon\}$ is compact for every ϵ).

References

[1] Babelon, O., Bernard, D. & Talon, M. *Introduction to Classical Integrable Systems.* (Cambridge University Press, 2003).